

Solutions – Practice Problem Set 1

ECON 101 – Macroeconomics

Fall 2025

1. Scarcity and Trade-offs

Scarcity means resources (time, income, natural resources) are limited relative to wants. Because of scarcity, choosing one option requires forgoing another (*trade-off*), and the *opportunity cost* of any choice is the value of the best alternative given up.

Examples.

- *Student time:* Spending three hours studying economics means forgoing three hours working for pay or relaxing. The opportunity cost of studying is the wage (or leisure) foregone.
- *Budget trade-off:* Buying a \$1,200 laptop today may force postponing a vacation. The opportunity cost of the laptop is the next-best use of that \$1,200 (e.g., part of tuition or the vacation).

Why fundamental? Every decision requires comparing marginal benefits with marginal opportunity costs, precisely because resources are scarce.

2. Economic Models

Why use models? Models simplify reality to isolate causal mechanisms (e.g., how price affects quantity demanded). Assumptions (*ceteris paribus*) help derive clear, testable predictions. They guide measurement and counterfactual analysis (“what if” policy experiments).

Strengths. Clarity, internal consistency, predictive discipline, and portability across contexts.

Limitations. Omitted variables/institutions, parameter instability (behavior may change with policy), measurement error, and potential mismatch between assumptions and context. Good practice: use models as *maps*, validated and refined with data.

3. Positive vs. Normative Analysis

Positive (descriptive/testable): “A \$10 billion increase in government spending raises real GDP in the short run.” This can be confronted with data.

Normative (prescriptive/value-laden): “The government *should* increase spending to reduce unemployment.” This depends on goals (equity vs. efficiency) and value judgments.

Why distinction matters. It separates evidence-based claims from value-based recommendations, clarifying where economists agree (facts/estimates) and where disagreement stems from different social objectives.

4. **Specialization and Trade (Comparative Advantage)**

Key logic: Even if one party is absolutely more productive in all goods, each can gain by specializing in the good with the *lower opportunity cost* and trading for the other good.

Steps: (i) Compute opportunity costs in each country/person. (ii) Assign production to the activity with lower opportunity cost. (iii) Choose a mutually beneficial terms of trade between the two opportunity costs. (iv) Both consume beyond their individual PPFs.

Intuition: Specialization aligns production with relative efficiencies, expanding total output; trade reallocates that surplus to improve both parties' consumption possibilities.

5. **PPF: Points Inside/On/Outside**

Inside the PPF: Feasible but inefficient (unused labor/capital or misallocation). *Example:* Producing 300 bushels of wheat and 60 cars when available resources could produce 400 bushels of wheat and 70 cars.

On the PPF: Feasible and productively efficient (cannot produce more of one good without less of the other). *Example:* A point consistent with full employment and best-known technology.

Outside the PPF: Currently unattainable with given resources/technology. *Example:* 900 bushels of wheat and 120 cars when the economy's maximum combinations fall short; attainable only with growth, trade, or technology improvements.

6. **Opportunity Cost Calculation**

Given: Essay takes 5 hours and yields benefit valued at \$80. Working pays \$15/hour.

(a) Opportunity cost of writing the essay.

$$\text{Wage foregone} = 5 \text{ hours} \times \$15/\text{hour} = \$75.$$

So the opportunity cost of writing the essay is \$75 (best alternative given up).

(b) Compare benefit (\$80) to opportunity cost (\$75). Net gain from essay:

$$\$80 - \$75 = \$5 > 0.$$

She should write the essay if she values the grade improvement at \$80 (and there are no other relevant costs/benefits). A sensitivity note: if her true valuation were $< \$75$, working would dominate.

7. PPF and Trade-offs

Comb.	Wheat	Cars
<i>A</i>	0	100
<i>B</i>	200	90
<i>C</i>	400	70
<i>D</i>	600	40
<i>E</i>	800	0

(a) The plot below places Wheat on the horizontal axis and Cars on the vertical axis. The five observed efficient points are connected with a smooth, concave envelope, reflecting increasing opportunity cost.

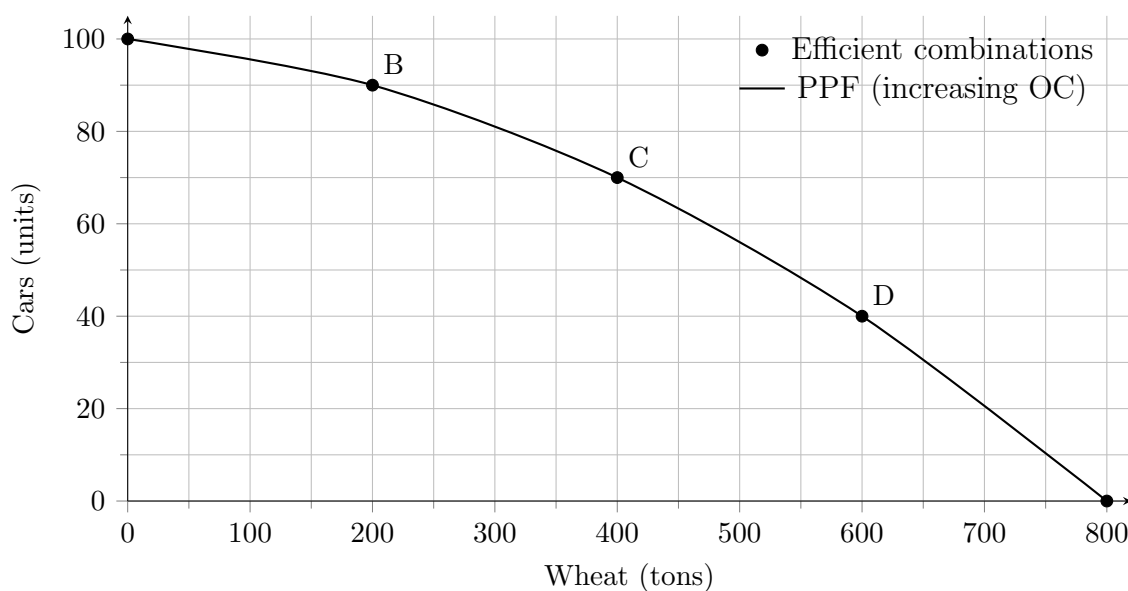


Figure 1. Production Possibilities Frontier for Wheat and Cars.

(b) **Opportunity cost from B to C.** Moving $B \rightarrow C$: Wheat increases by +200, Cars decrease by -20 .

$$\text{OC of } +200 \text{ wheat} = 20 \text{ cars} \Rightarrow 0.10 \text{ car per ton of wheat.}$$

Equivalently, 1 car costs 10 tons of wheat at this margin.

(c) **Is OC increasing?** Along each segment:

$$A \rightarrow B : 0.05 \text{ car/ton} \quad B \rightarrow C : 0.10 \text{ car/ton} \quad C \rightarrow D : 0.15 \text{ car/ton} \quad D \rightarrow E : 0.20 \text{ car/ton}$$

Opportunity cost rises monotonically as wheat output expands: **increasing** OC.

8. Comparative Advantage and Trade

	Cloth	Wine
Alpha	60	30
Beta	40	20

Assume linear PPFs (constant OC).

(a) Opportunity costs.

Alpha: 1 cloth = 0.5 wine; 1 wine = 2 cloth.

Beta: 1 cloth = 0.5 wine; 1 wine = 2 cloth.

(b) Comparative advantage. Identical OCs $\Rightarrow$ **neither** country has a comparative advantage.

(c) Gains from trade? With identical OCs and CRS, *no specialization gains* arise at terms of trade equal to the common OC.